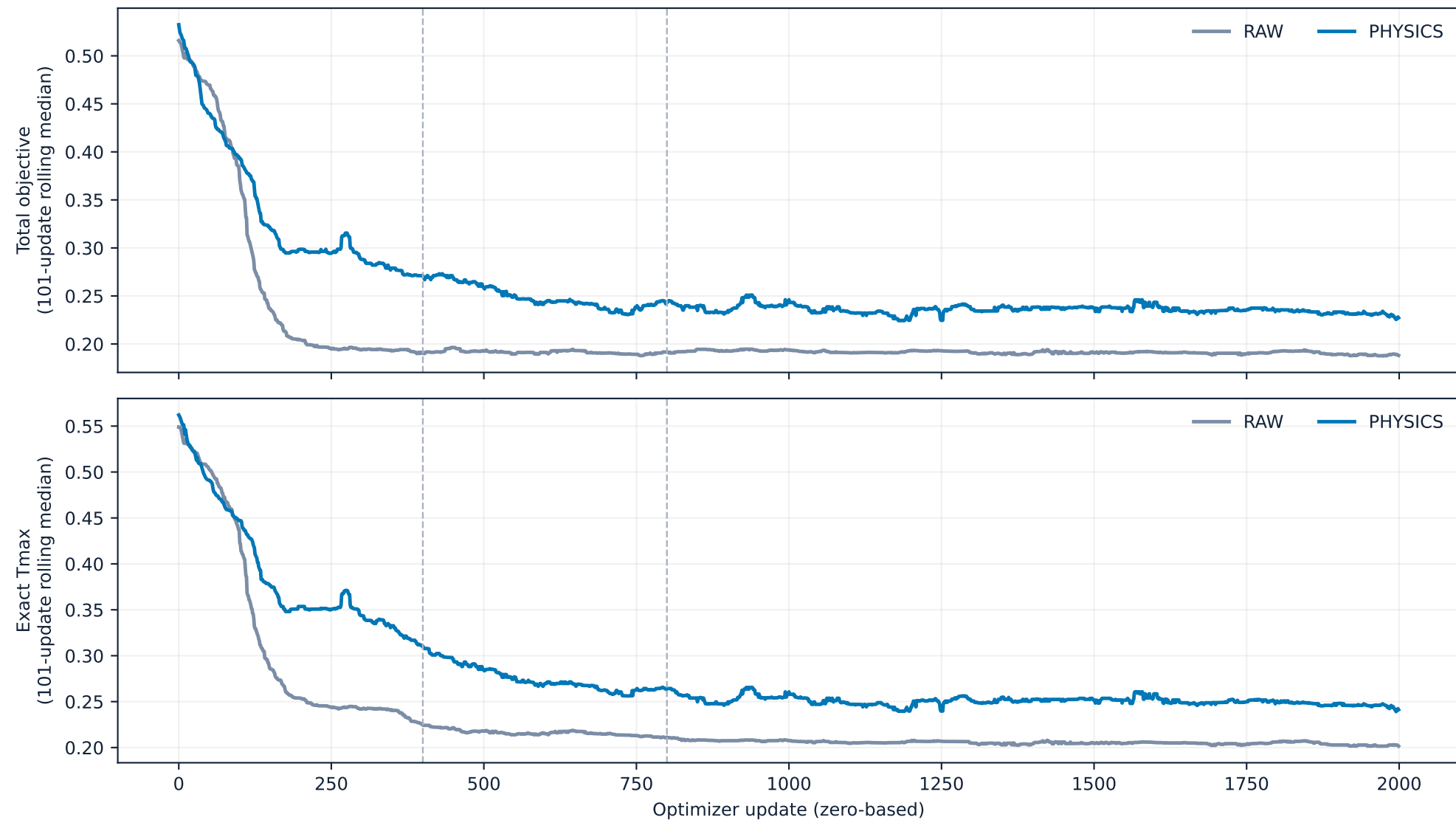


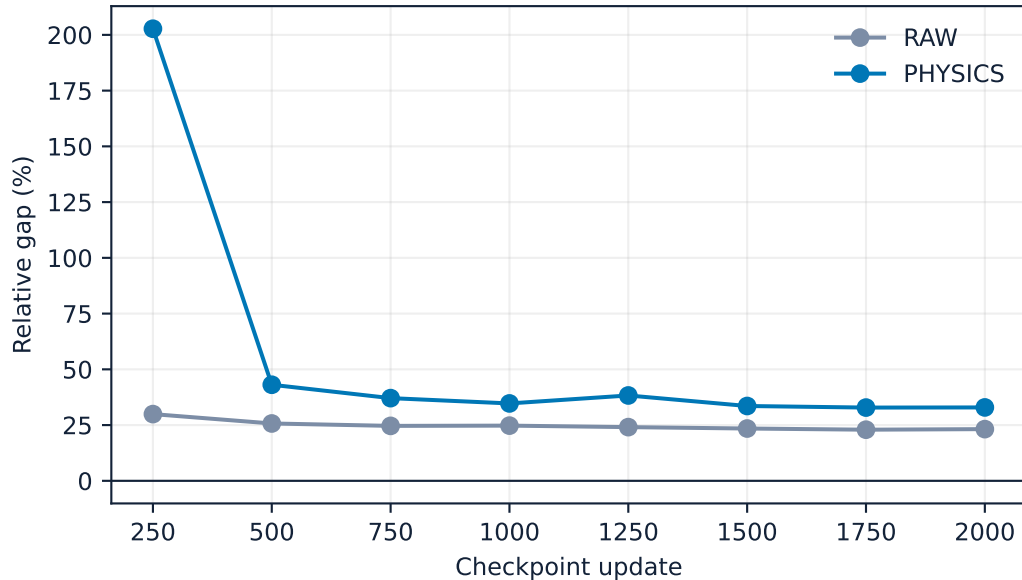
Matched RAW vs physics-transformed training

Same initialization, task stream, optimizer and NCA; only conditioning representation differs

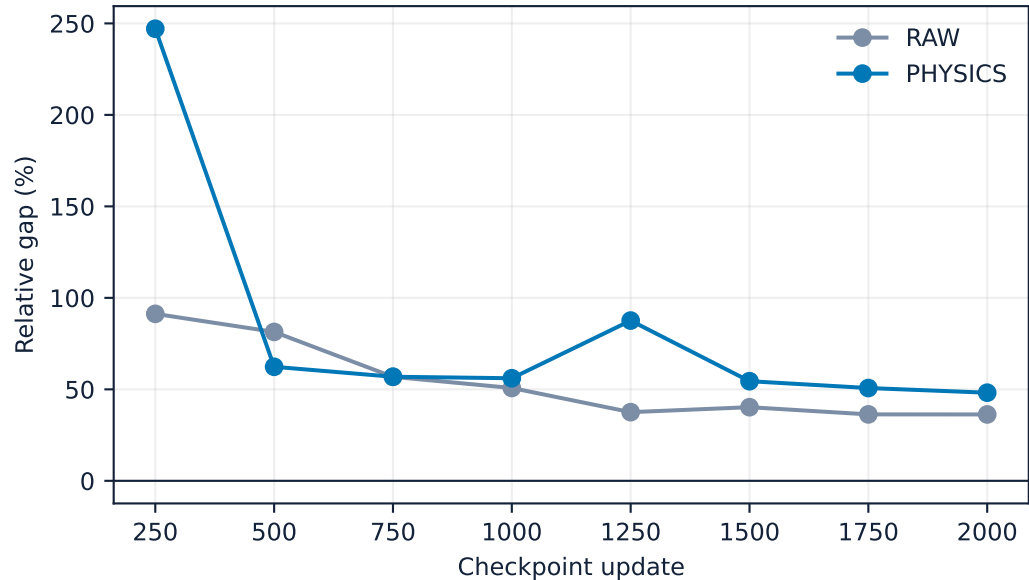


Frozen-checkpoint quality on all 32 validation layouts

Median gap to 600-step gradient

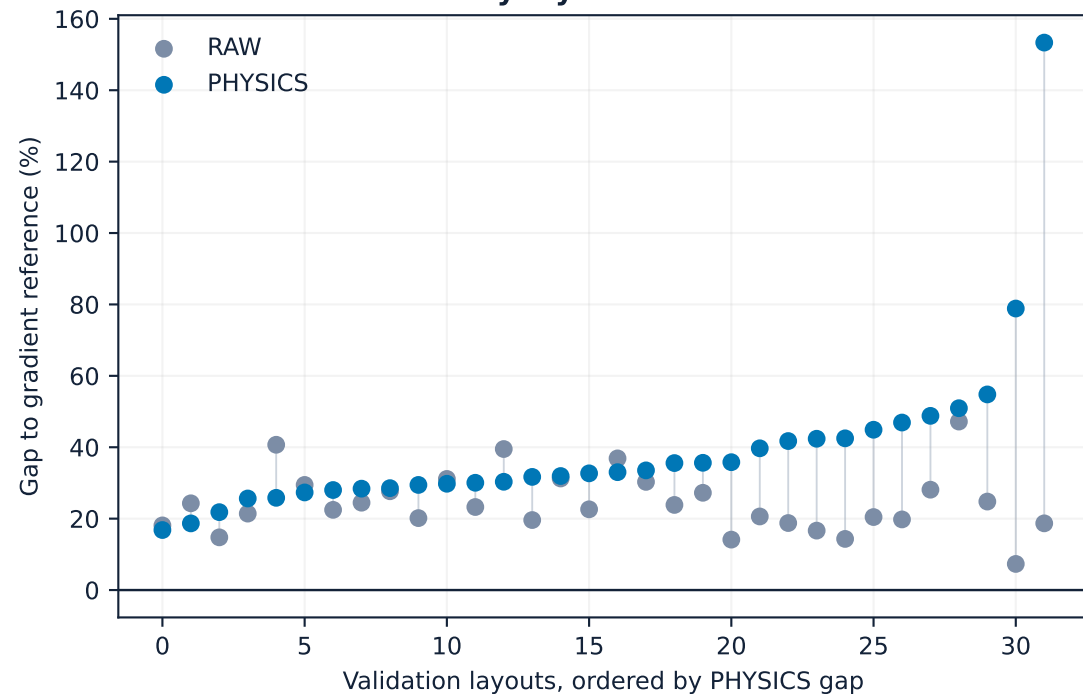


90th-percentile gap

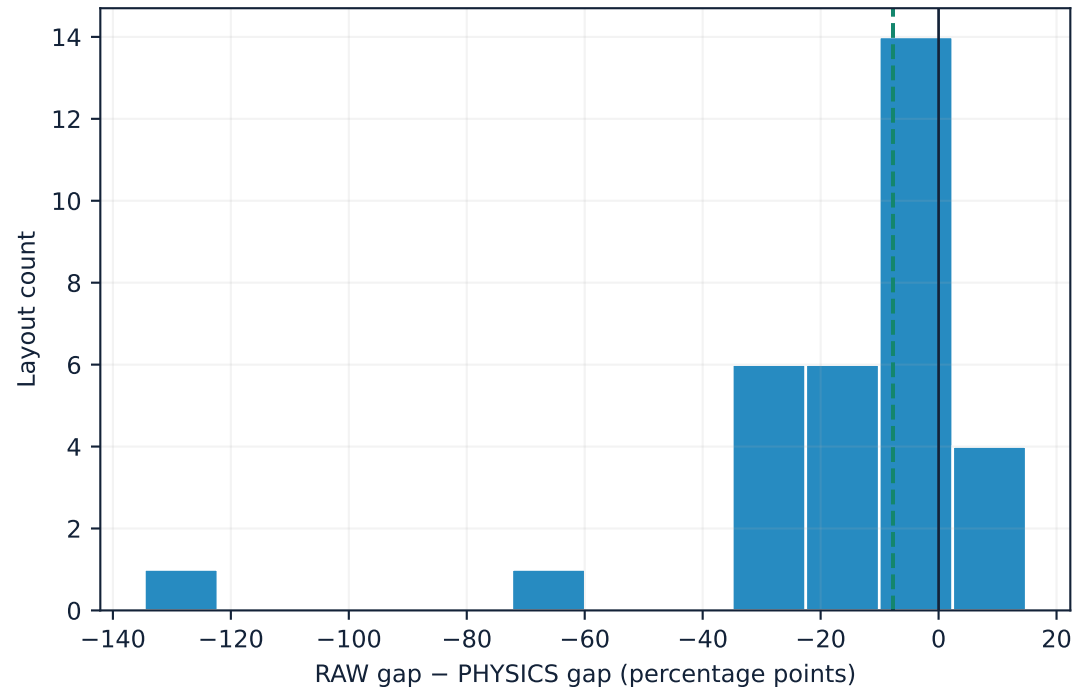


Paired conditioning effect on unseen validation layouts

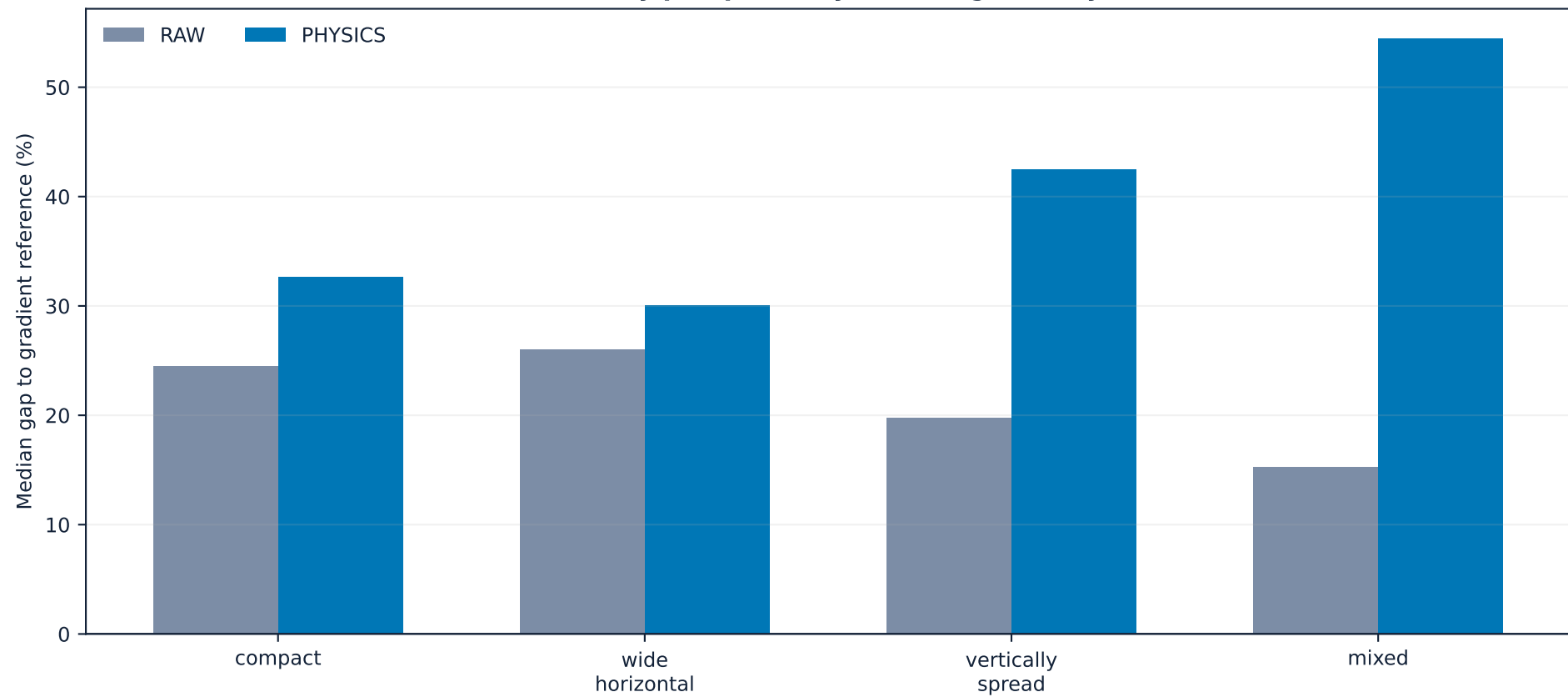
Every layout is shown



Paired improvement; median=-7.73 pp



Performance by prospectively defined geometry stratum



Disclosed illustrative ranks: best, median-nearest and worst PHYSICS gap

600-step gradient

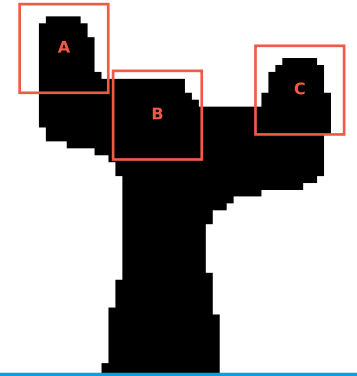
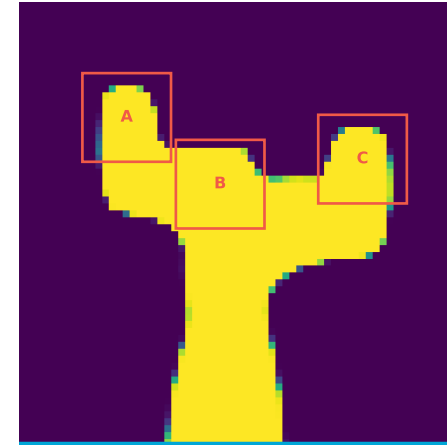
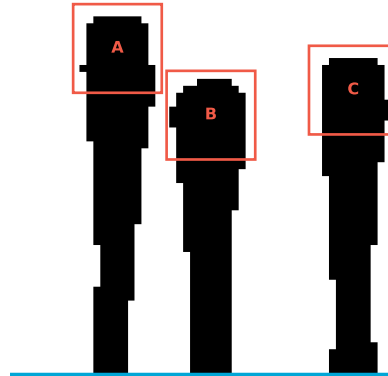
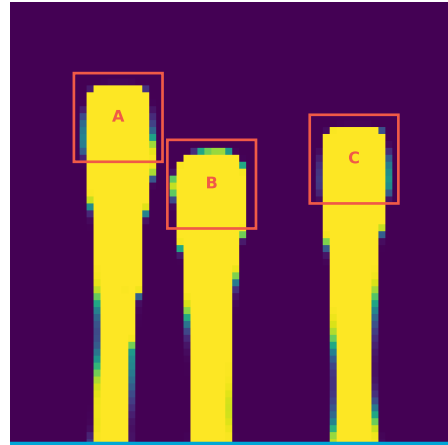
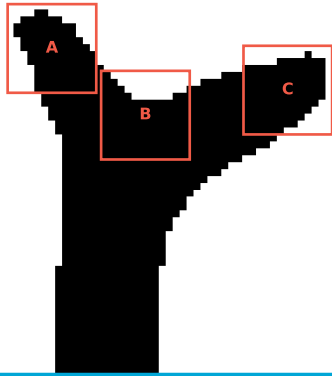
RAW continuous

RAW binary

PHYSICS continuous

PHYSICS binary

BEST rank
layout 24
PHYSICS gap 16.81%



600-step gradient

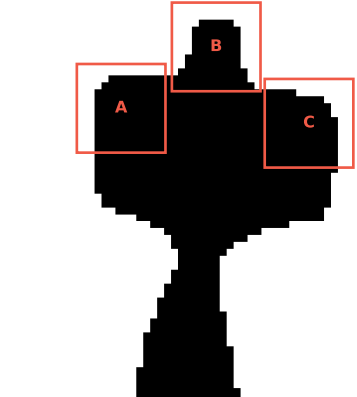
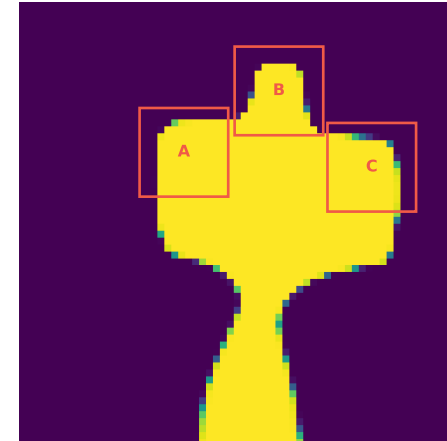
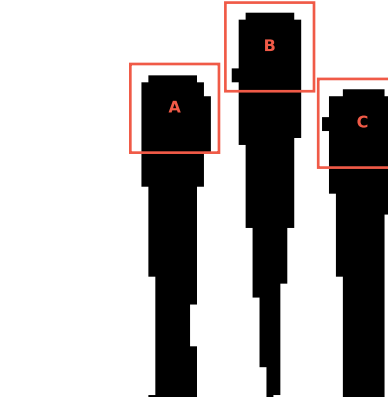
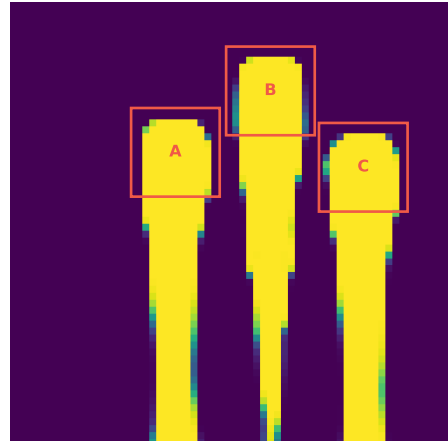
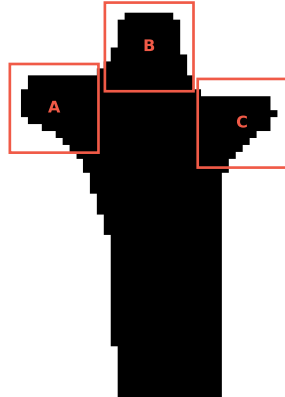
RAW continuous

RAW binary

PHYSICS continuous

PHYSICS binary

MEDIAN rank
layout 03
PHYSICS gap 32.69%



600-step gradient

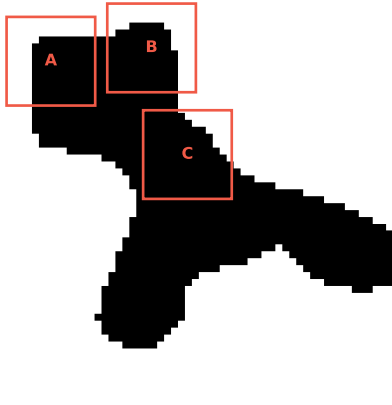
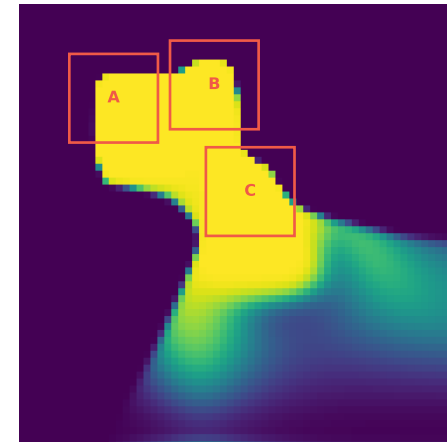
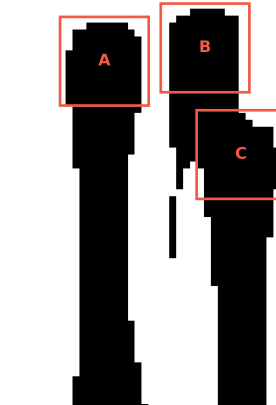
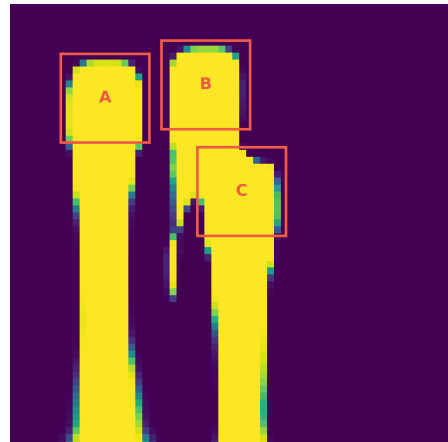
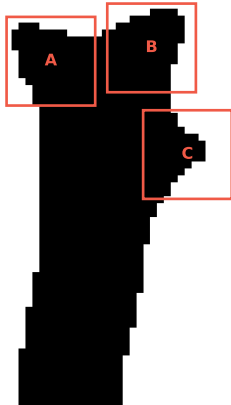
RAW continuous

RAW binary

PHYSICS continuous

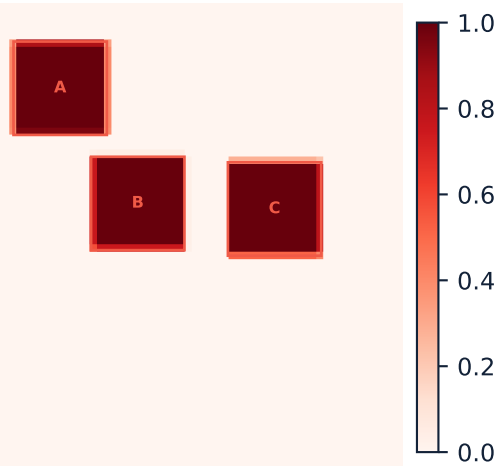
PHYSICS binary

WORST rank
layout 19
PHYSICS gap 153.34%

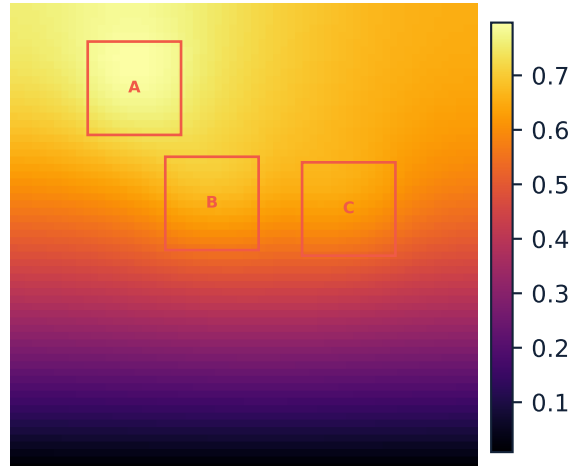


Physics-transformed conditioning contains no optimized topology or teacher design

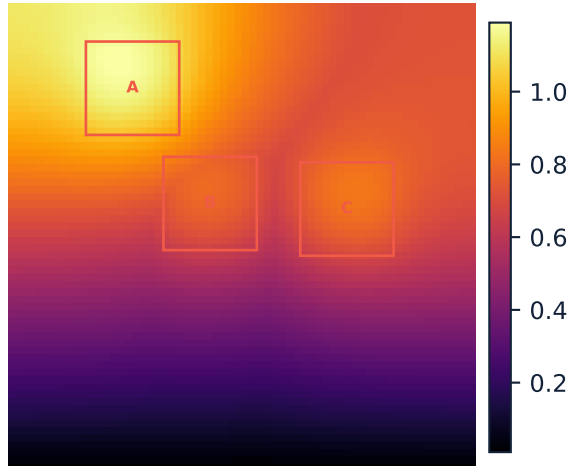
Source sum / 25



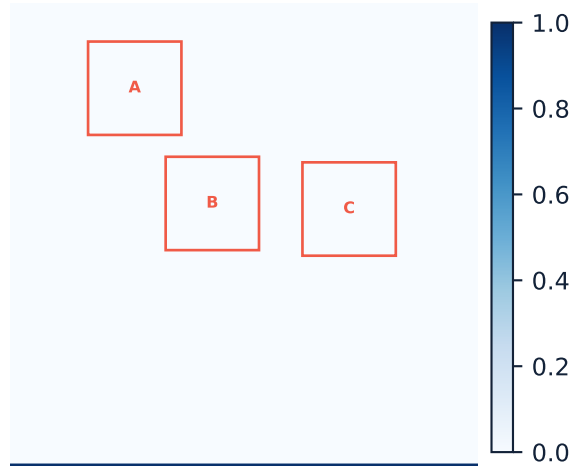
Low-k temperature mean



Low-k temperature max

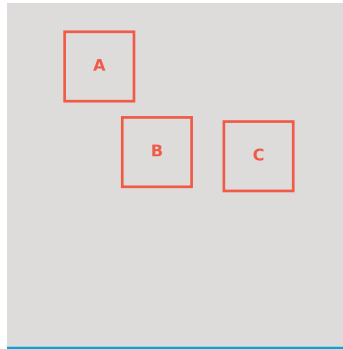


Sink mask

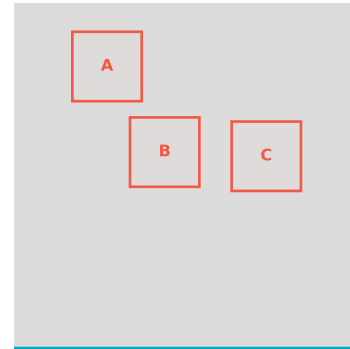


Frozen PHYSICS-NCA local growth from exact zero state • validation layout 00

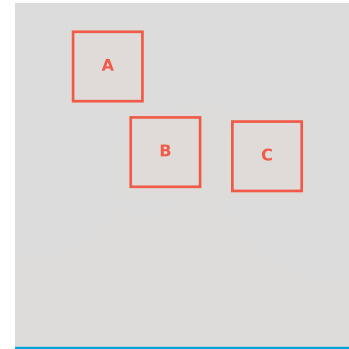
step 0



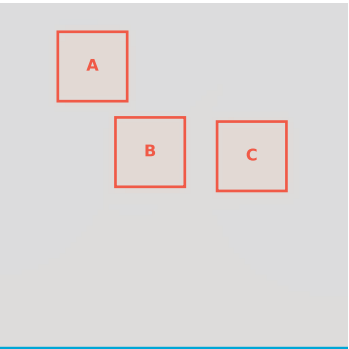
step 1



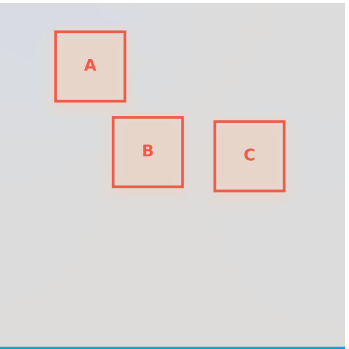
step 2



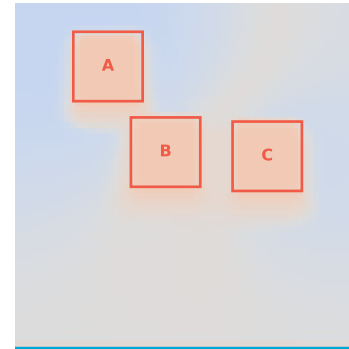
step 4



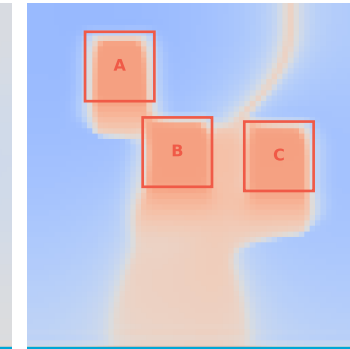
step 8



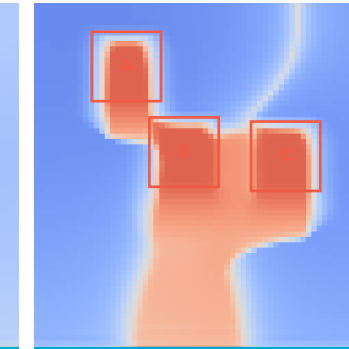
step 16



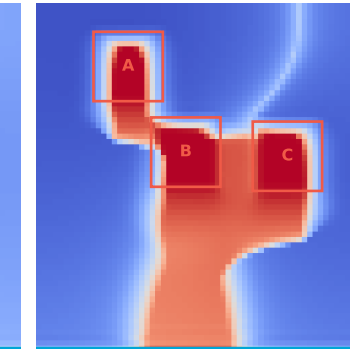
step 32



step 48

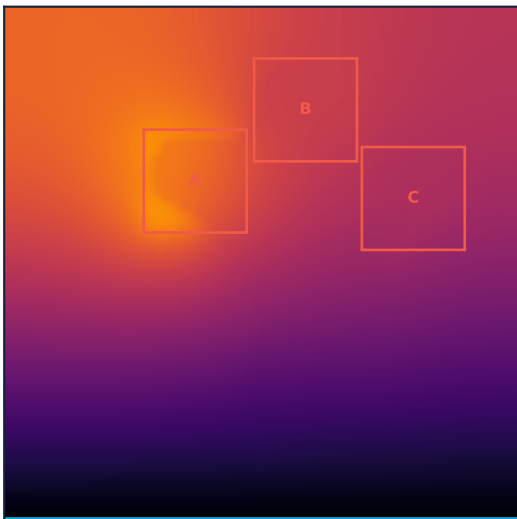


step 64

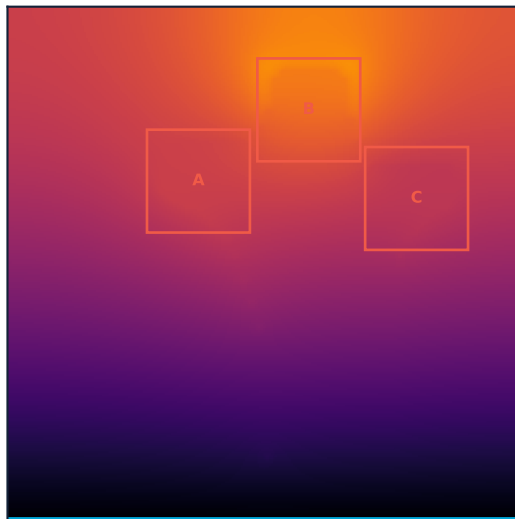


Independent SciPy 256×256 temperature fields • median-nearest layout 03

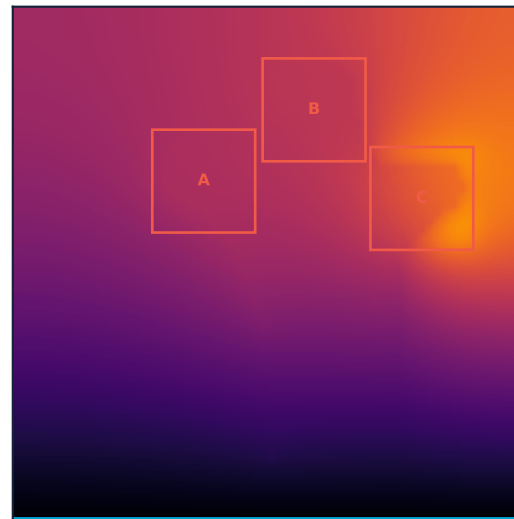
600-step gradient • A
Tmax=0.150818



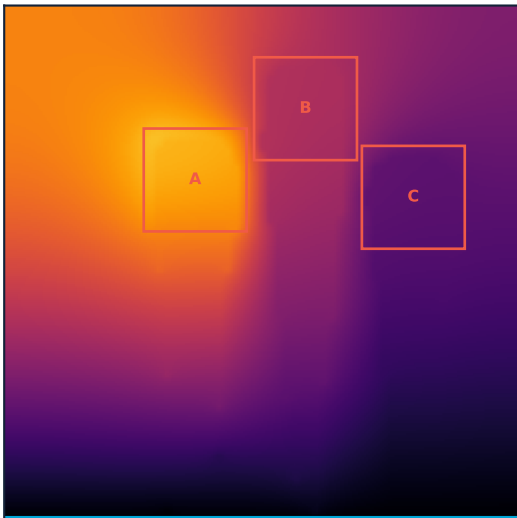
600-step gradient • B
Tmax=0.150138



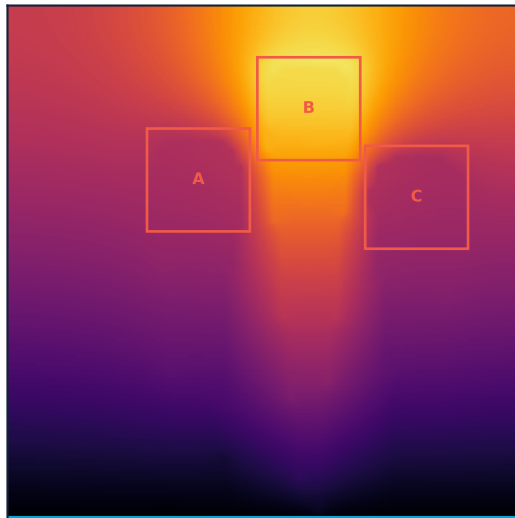
600-step gradient • C
Tmax=0.150050



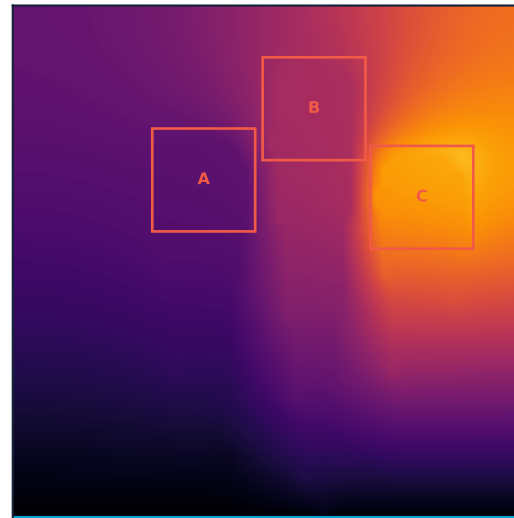
Frozen RAW-NCA • A
Tmax=0.169950



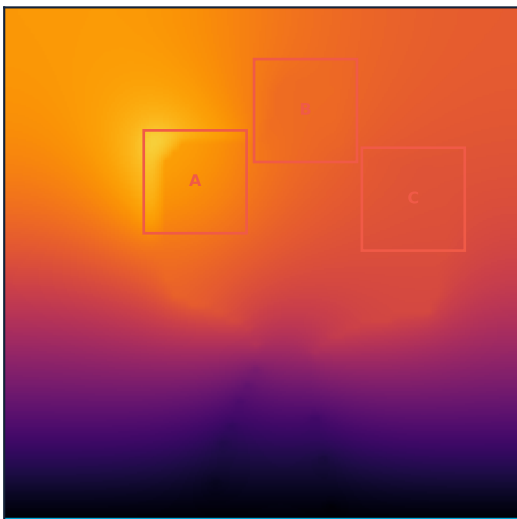
Frozen RAW-NCA • B
Tmax=0.184680



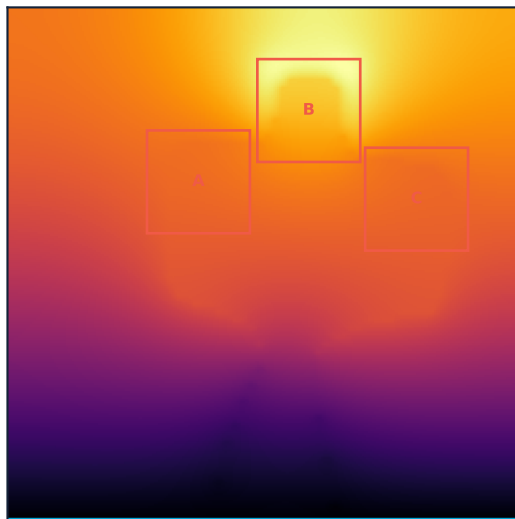
Frozen RAW-NCA • C
Tmax=0.167117



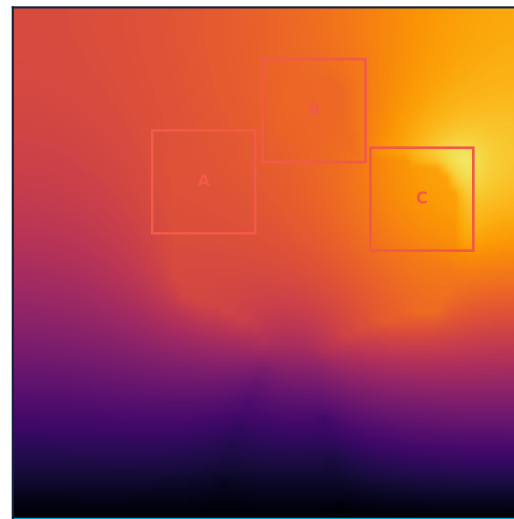
Frozen PHYSICS-NCA • A
Tmax=0.176013



Frozen PHYSICS-NCA • B
Tmax=0.199016



Frozen PHYSICS-NCA • C
Tmax=0.185258



Temperature

0.175

0.150

0.125

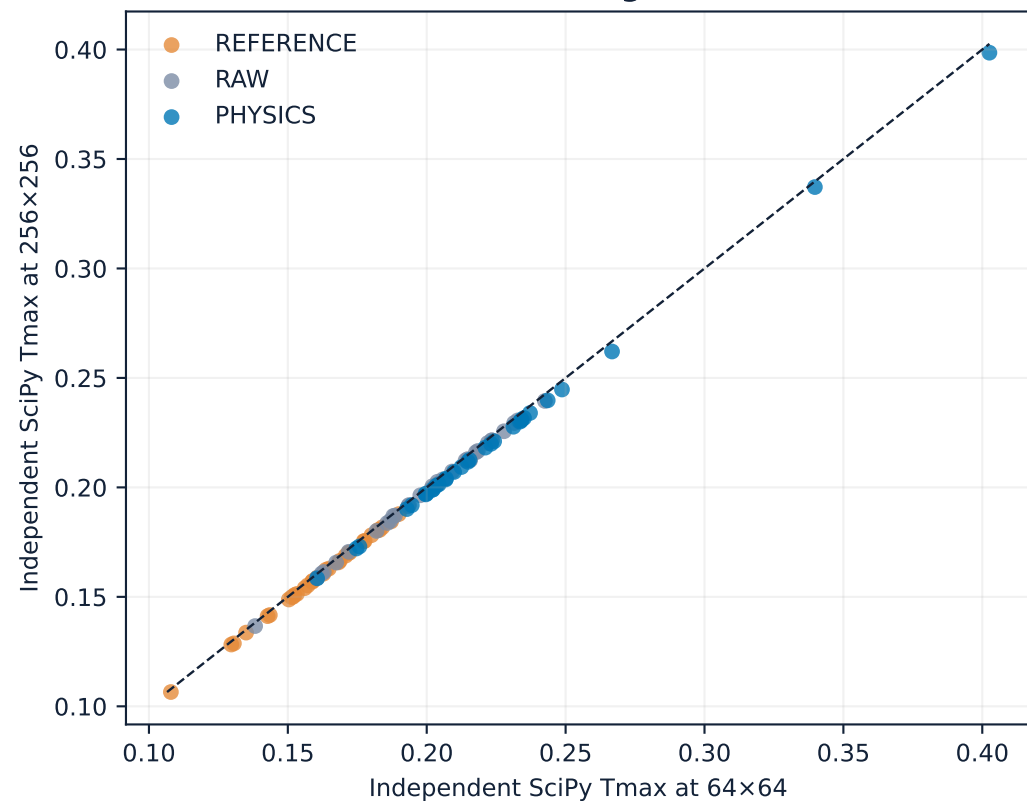
0.100

0.075

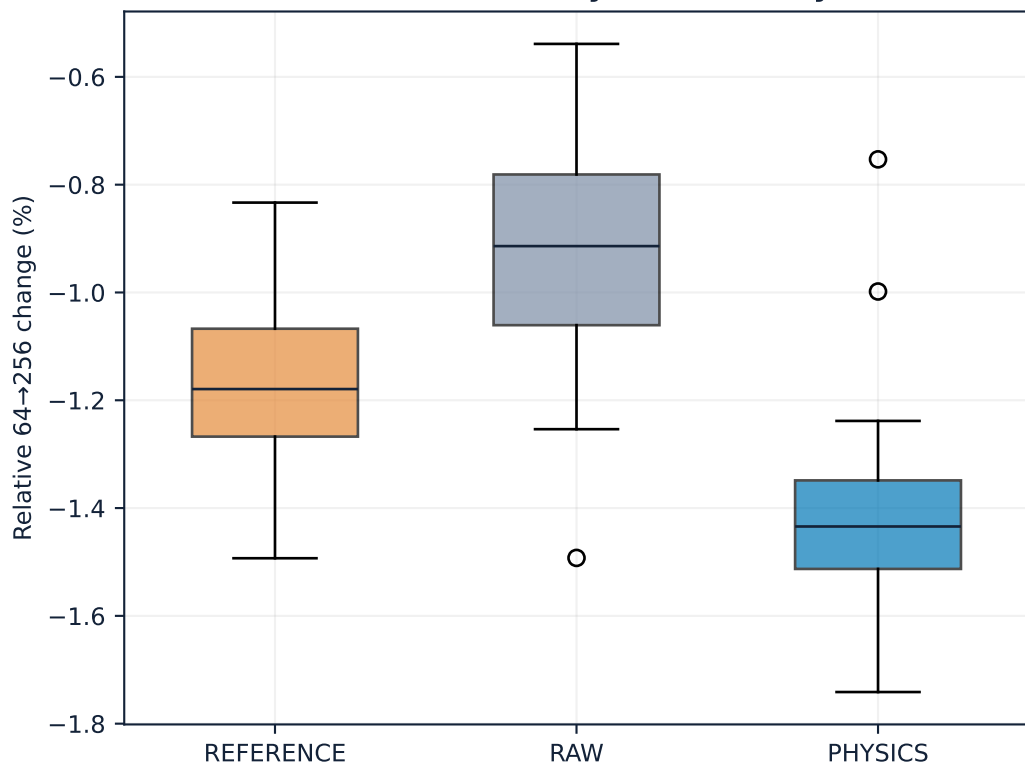
0.050

0.025

Grid-transfer agreement



Resolution sensitivity across all layouts



NCA-MT2B verdict: PHYSICS_NO_GO

paired conditioning-effect evidence did not pass every locked gate

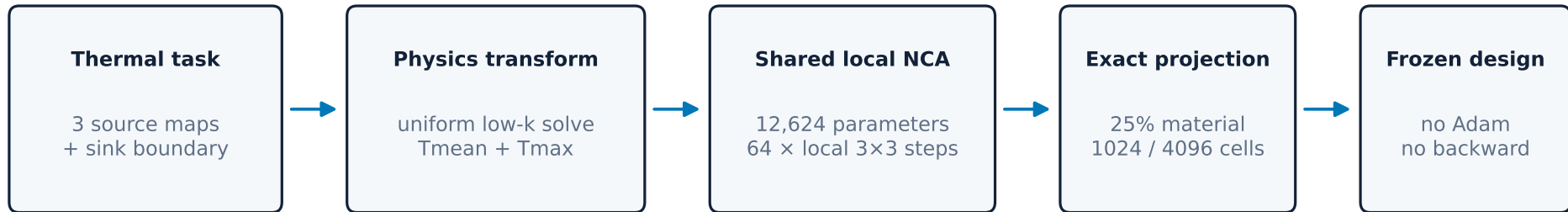
Frozen generator	Median gap	P90 gap	Wins vs gradient
RAW control	22.93%	36.32%	0/32
PHYSICS	32.85%	50.72%	0/32

PHYSICS improves the paired gap on 7/32 layouts; median reduction -7.73 percentage points.

Paired bootstrap 95% CI for the median reduction: [-19.09, -3.74] percentage points.

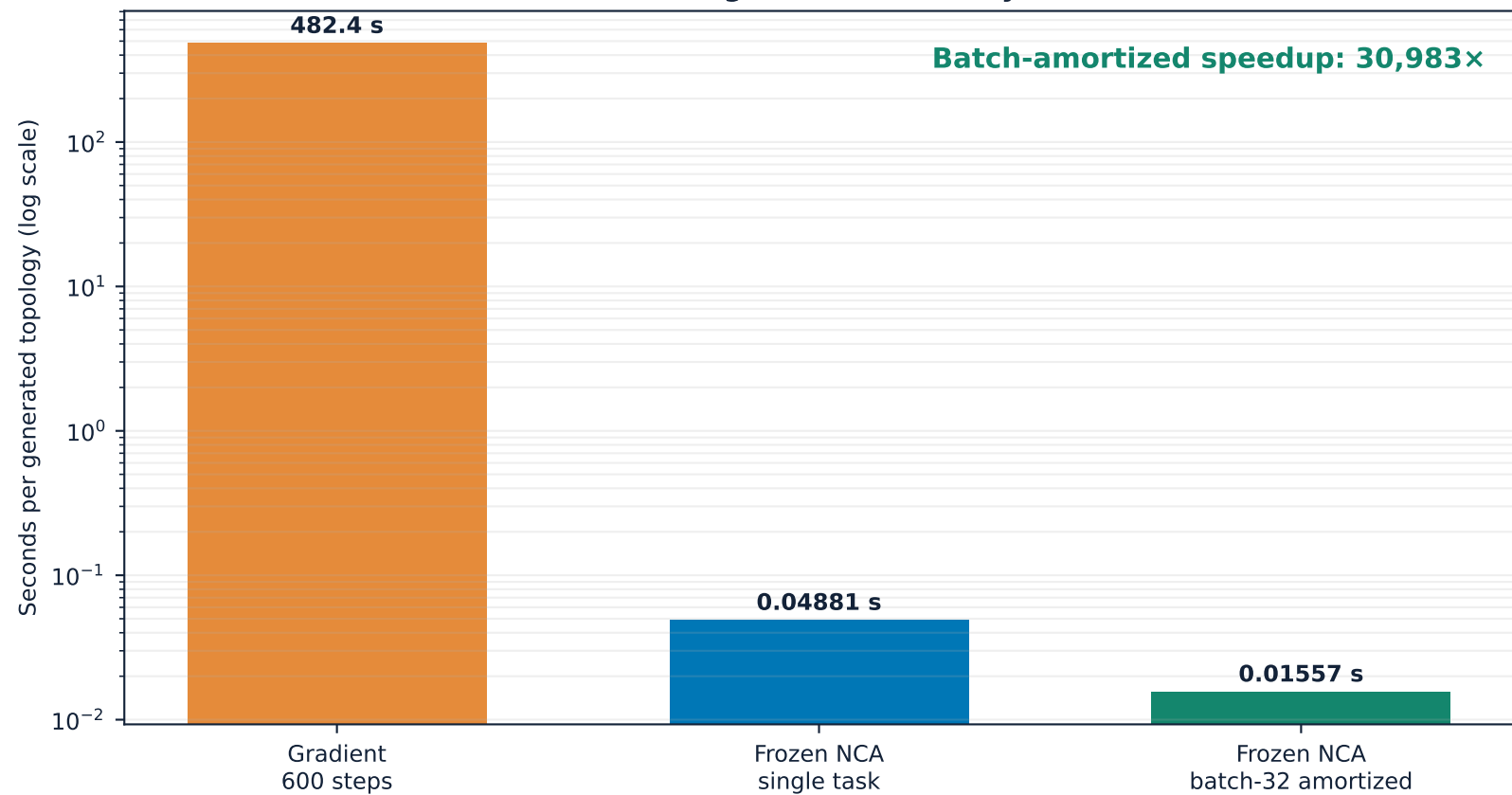
Scope: development validation ablation only. Test ID/OOD remain sealed; no claim of final generalization yet.

TRAINING: physics loss → backward through differentiable solver → same shared NCA weights



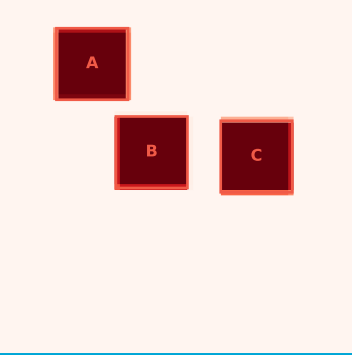
INFERENCE: one unseen layout → fixed physical representation → 64 local updates → topology

Conservative end-to-end generation latency on the same A100

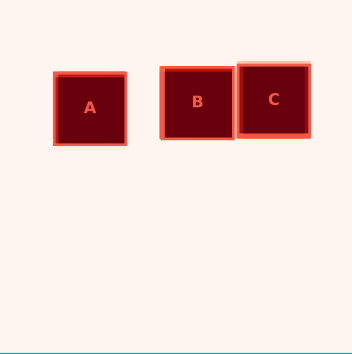


Complete validation topology atlas • layouts 00-07

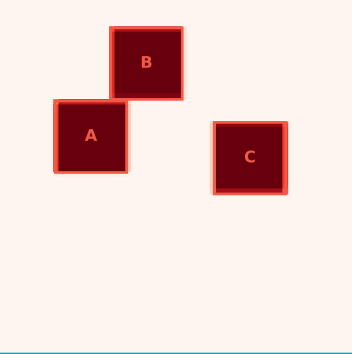
Layout 00 • vertically spread



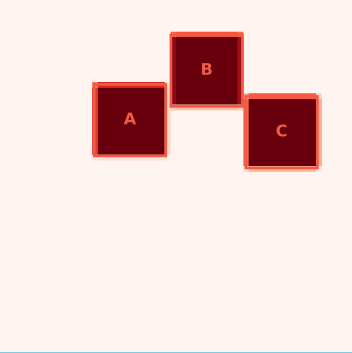
Layout 01 • wide horizontal



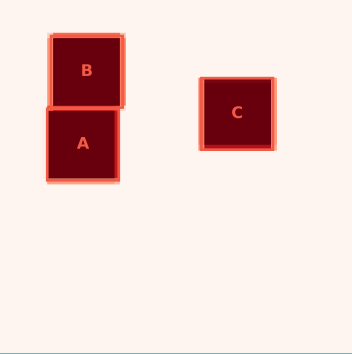
Layout 02 • vertically spread



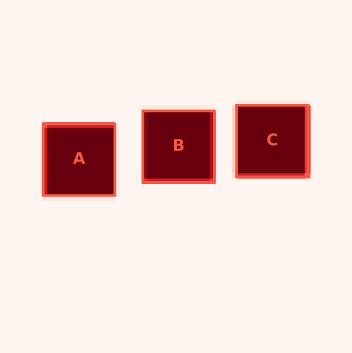
Layout 03 • compact



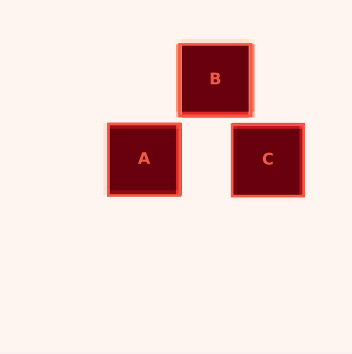
Layout 04 • compact



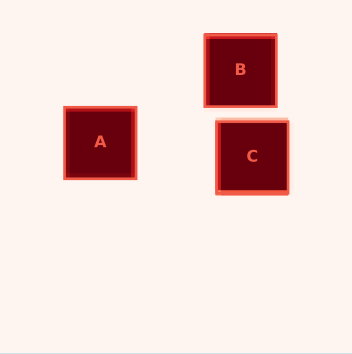
Layout 05 • wide horizontal



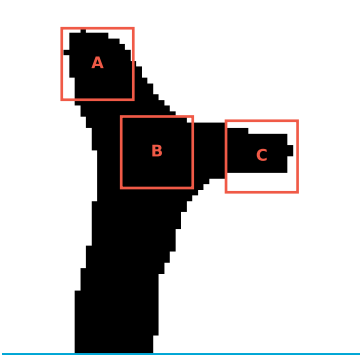
Layout 06 • vertically spread



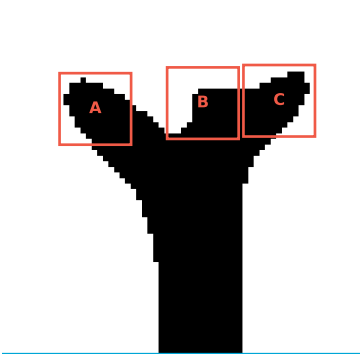
Layout 07 • vertically spread



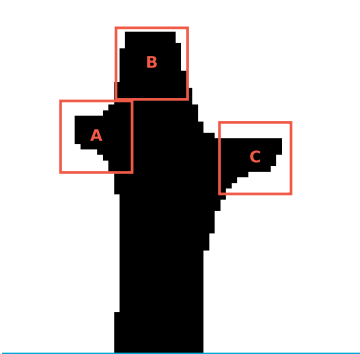
Gradient • Tmax=0.1718



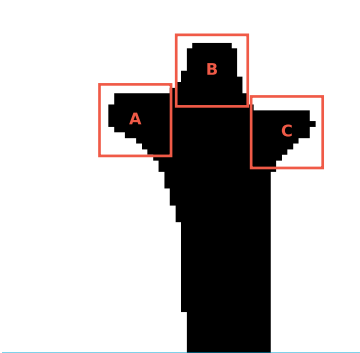
Gradient • Tmax=0.1687



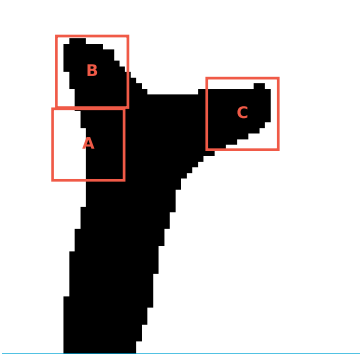
Gradient • Tmax=0.1519



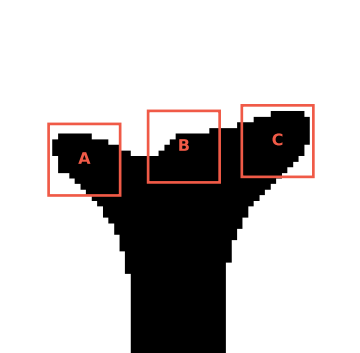
Gradient • Tmax=0.1524



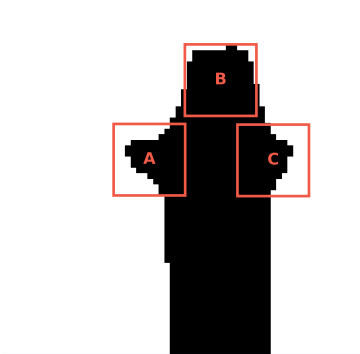
Gradient • Tmax=0.1723



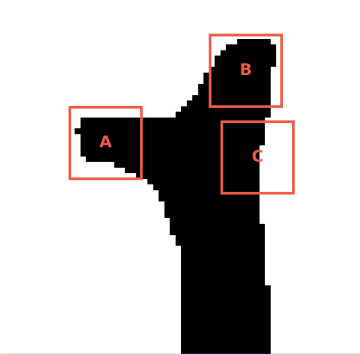
Gradient • Tmax=0.1306



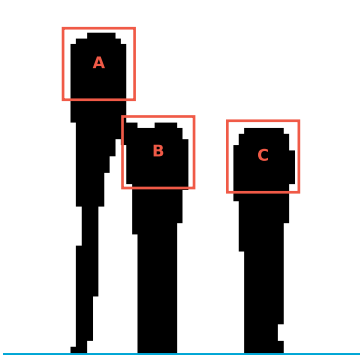
Gradient • Tmax=0.1297



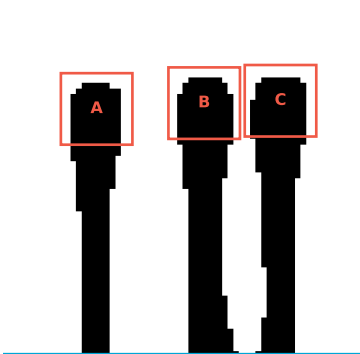
Gradient • Tmax=0.1571



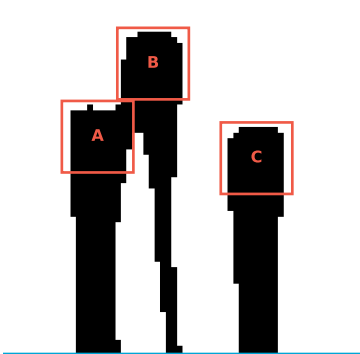
RAW • gap=18.8%



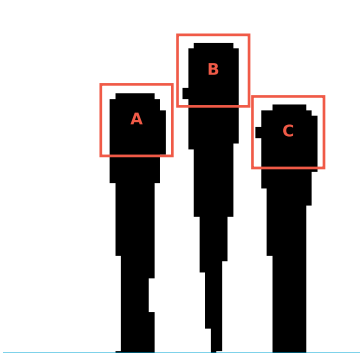
RAW • gap=29.4%



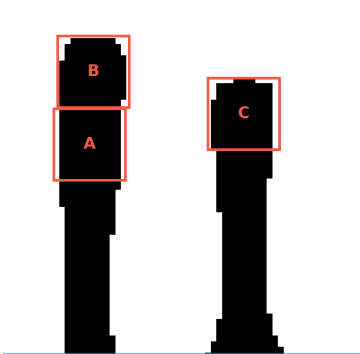
RAW • gap=19.8%



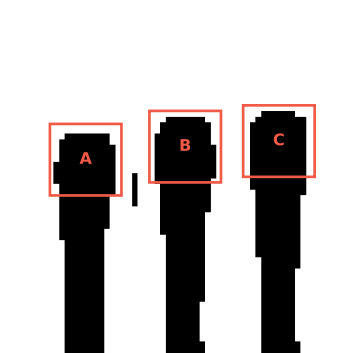
RAW • gap=22.6%



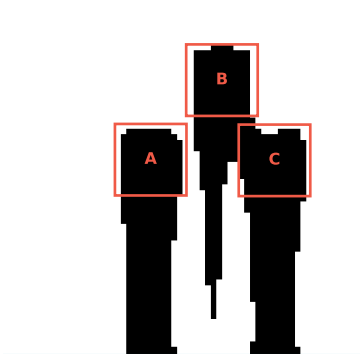
RAW • gap=14.8%



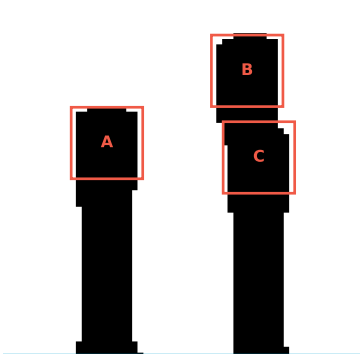
RAW • gap=24.8%



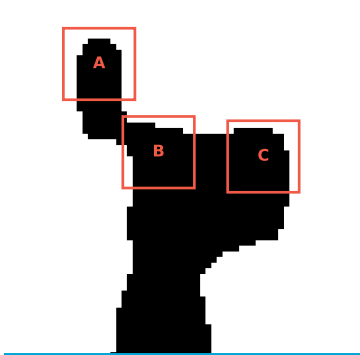
RAW • gap=23.8%



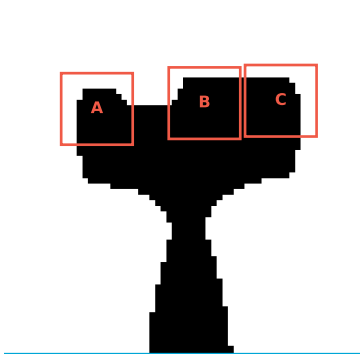
RAW • gap=19.6%



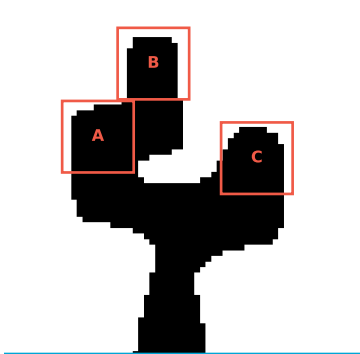
PHYSICS • gap=41.7%



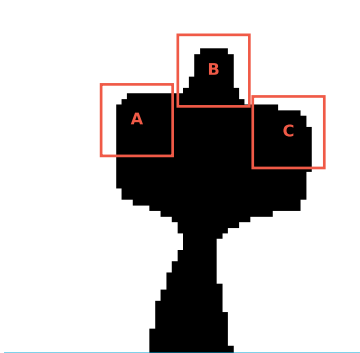
PHYSICS • gap=27.3%



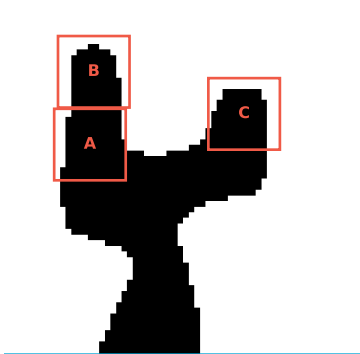
PHYSICS • gap=46.9%



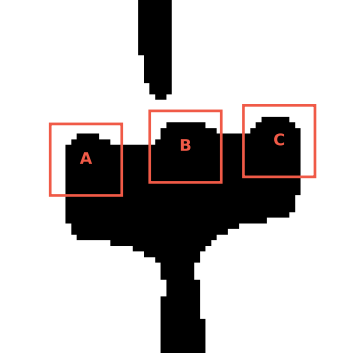
PHYSICS • gap=32.7%



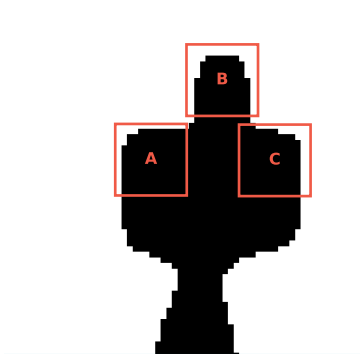
PHYSICS • gap=21.8%



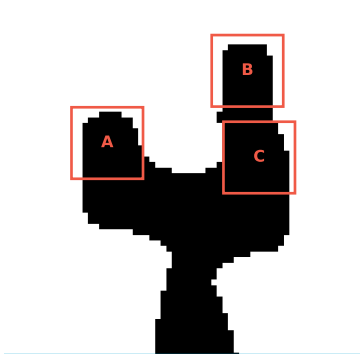
PHYSICS • gap=54.8%



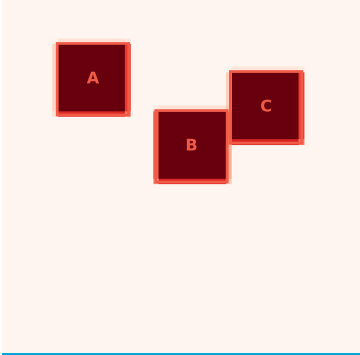
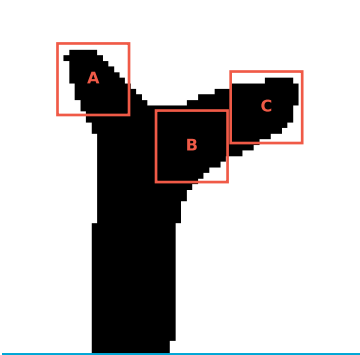
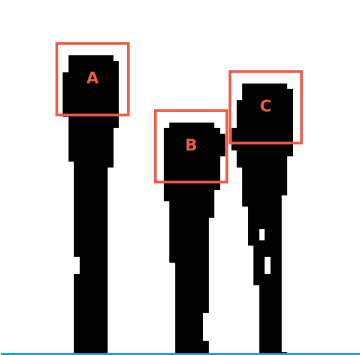
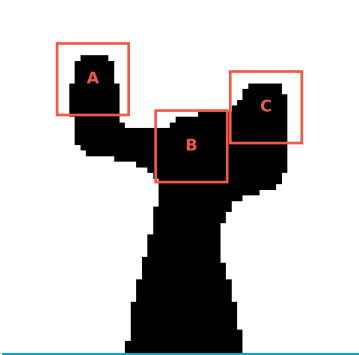
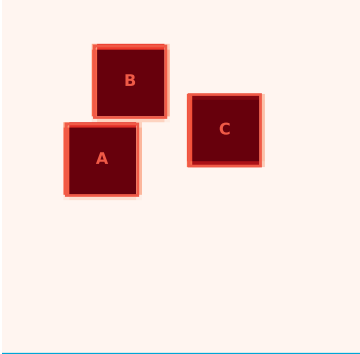
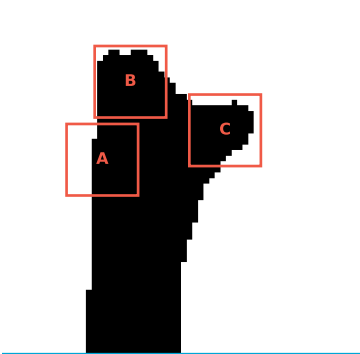
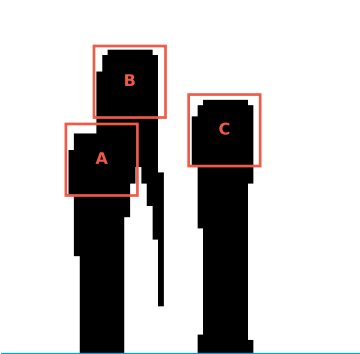
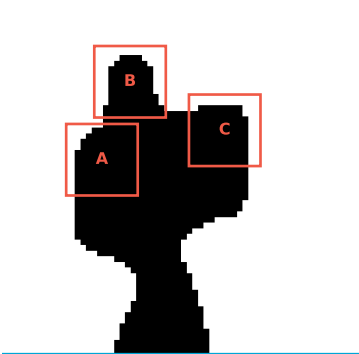
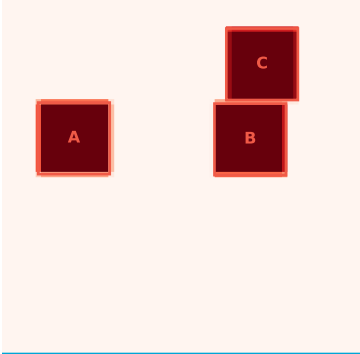
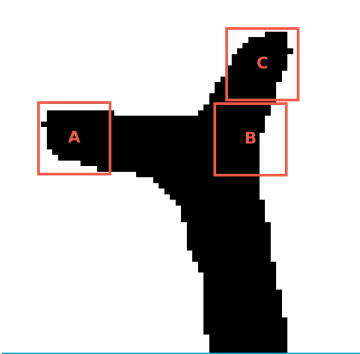
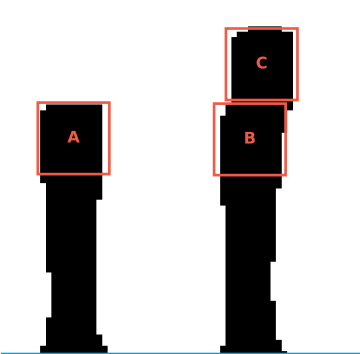
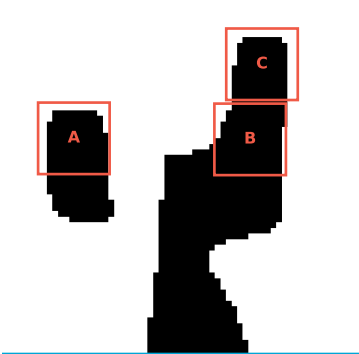
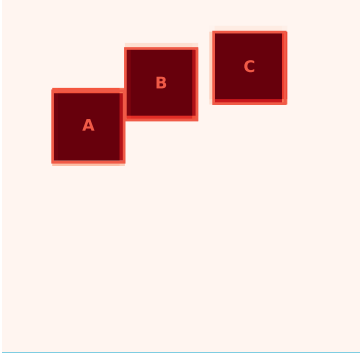
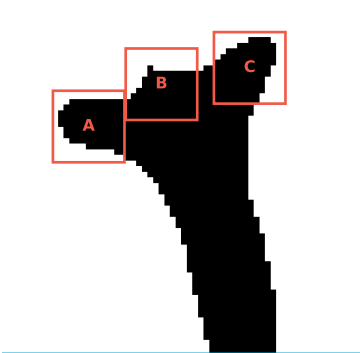
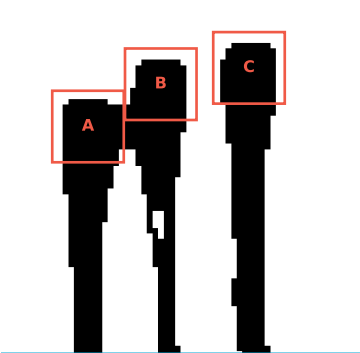
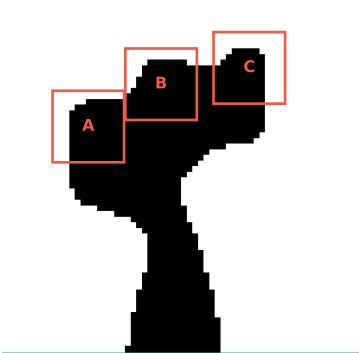
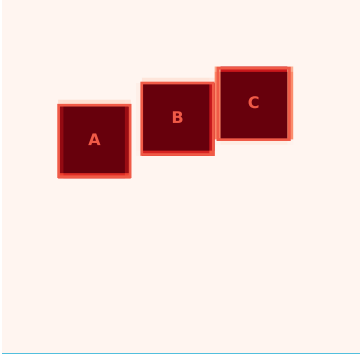
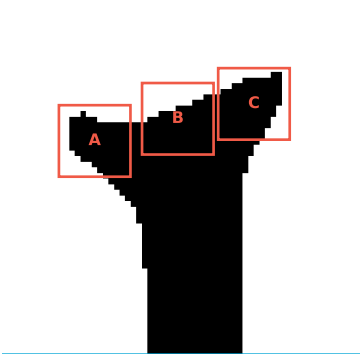
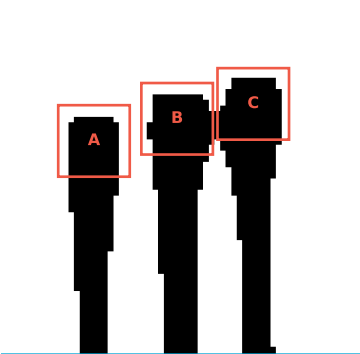
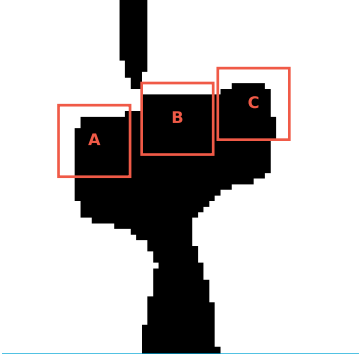
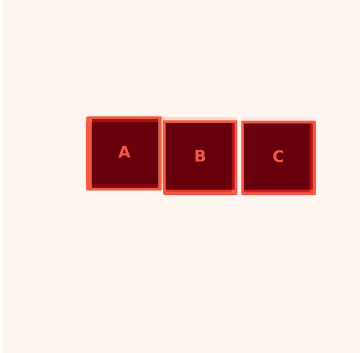
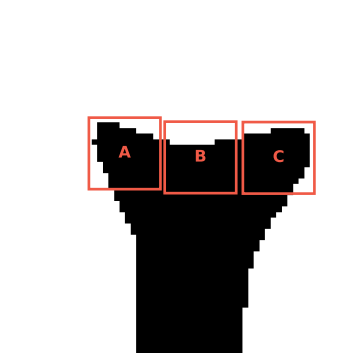
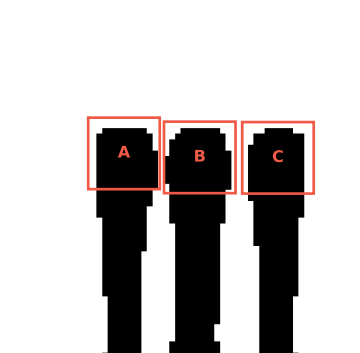
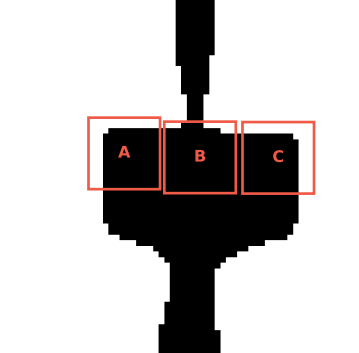
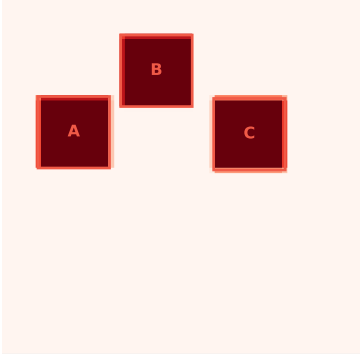
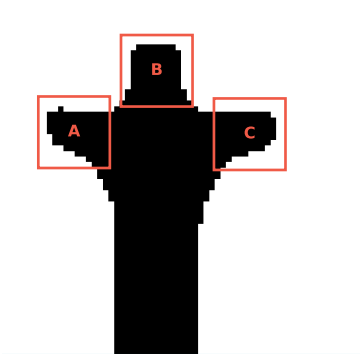
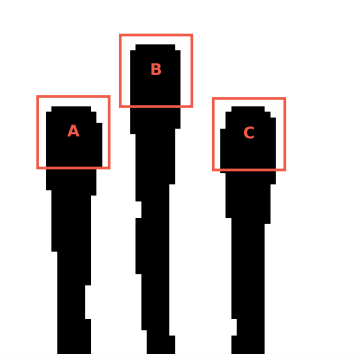
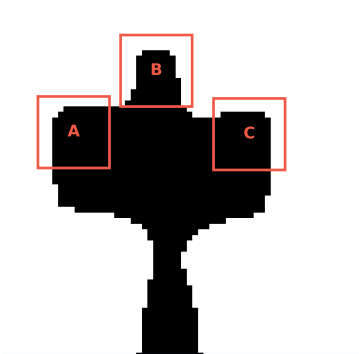
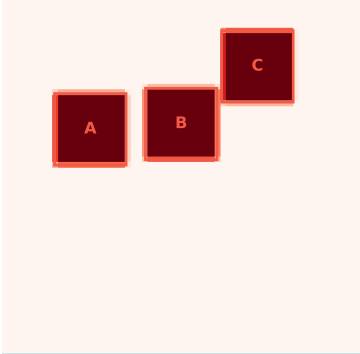
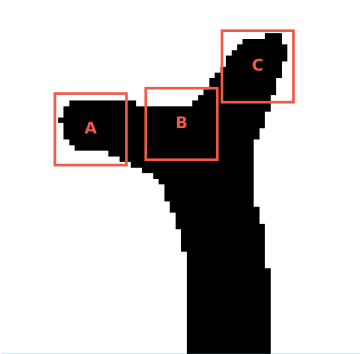
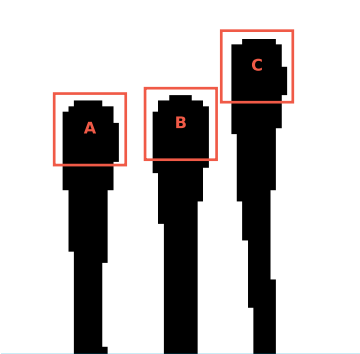
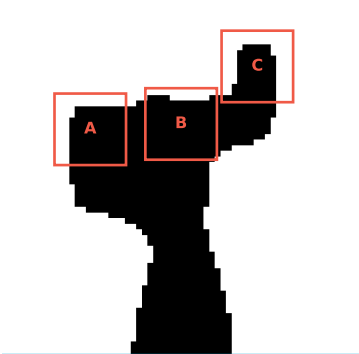
PHYSICS • gap=35.6%



PHYSICS • gap=31.7%

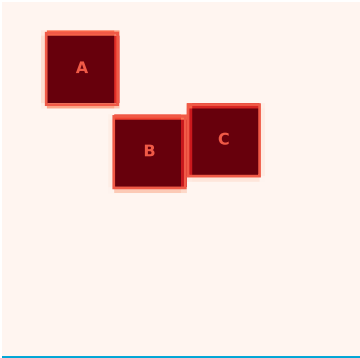


Complete validation topology atlas • layouts 08-15

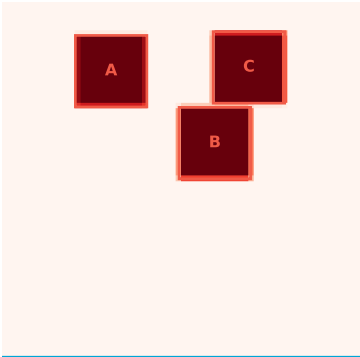
Layout 08 • wide horizontal 	Gradient • Tmax=0.1776 	RAW • gap=30.3% 	PHYSICS • gap=33.5% 
Layout 09 • vertically spread 	Gradient • Tmax=0.1350 	RAW • gap=20.2% 	PHYSICS • gap=29.4% 
Layout 10 • mixed 	Gradient • Tmax=0.1900 	RAW • gap=7.3% 	PHYSICS • gap=78.9% 
Layout 11 • compact 	Gradient • Tmax=0.1830 	RAW • gap=24.5% 	PHYSICS • gap=28.4% 
Layout 12 • compact 	Gradient • Tmax=0.1435 	RAW • gap=16.6% 	PHYSICS • gap=42.4% 
Layout 13 • compact 	Gradient • Tmax=0.1079 	RAW • gap=28.1% 	PHYSICS • gap=48.8% 
Layout 14 • wide horizontal 	Gradient • Tmax=0.1605 	RAW • gap=20.6% 	PHYSICS • gap=39.7% 
Layout 15 • wide horizontal 	Gradient • Tmax=0.1773 	RAW • gap=31.3% 	PHYSICS • gap=31.9% 

Complete validation topology atlas • layouts 16-23

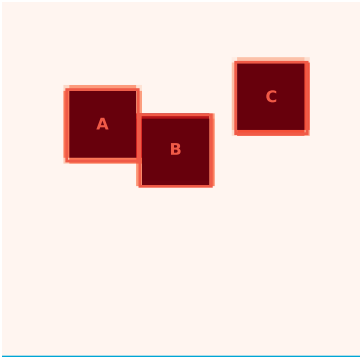
Layout 16 • vertically spread



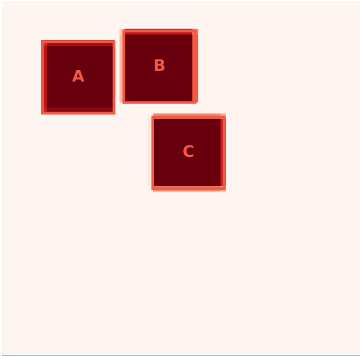
Layout 17 • vertically spread



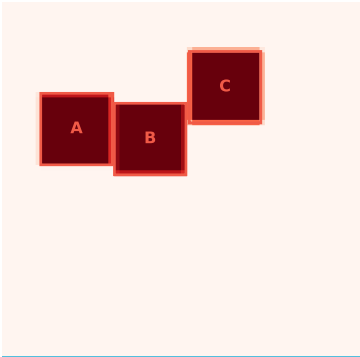
Layout 18 • wide horizontal



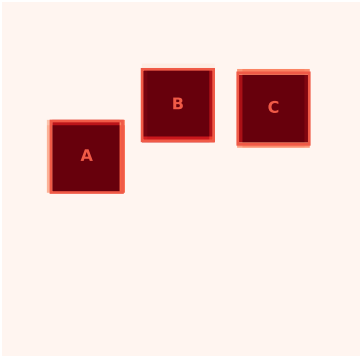
Layout 19 • vertically spread



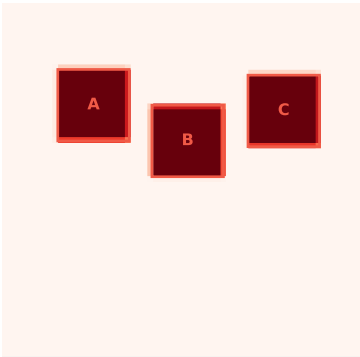
Layout 20 • compact



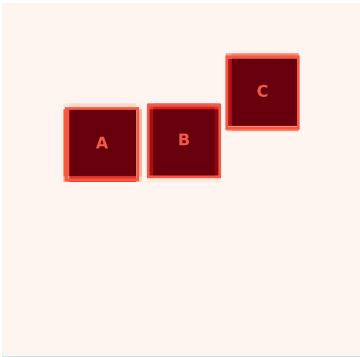
Layout 21 • wide horizontal



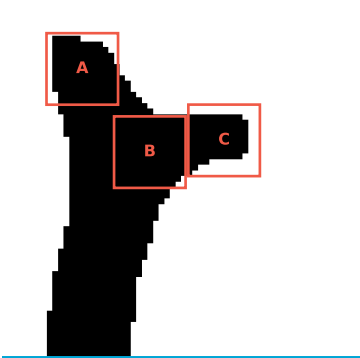
Layout 22 • wide horizontal



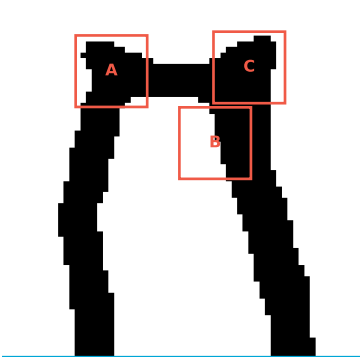
Layout 23 • compact



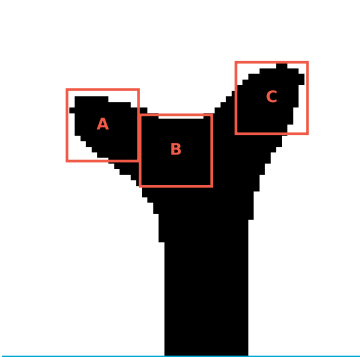
Gradient • Tmax=0.1648



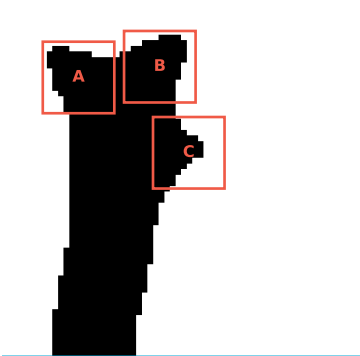
Gradient • Tmax=0.1872



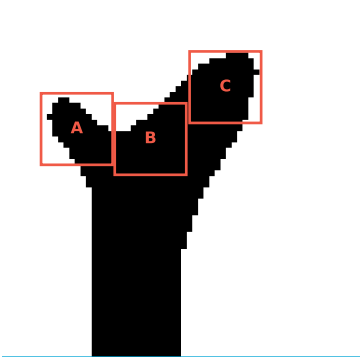
Gradient • Tmax=0.1561



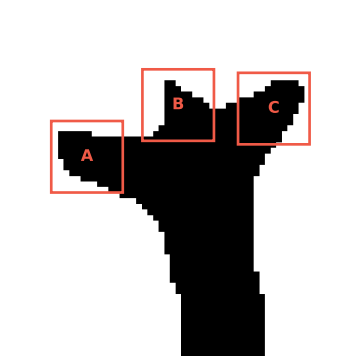
Gradient • Tmax=0.1589



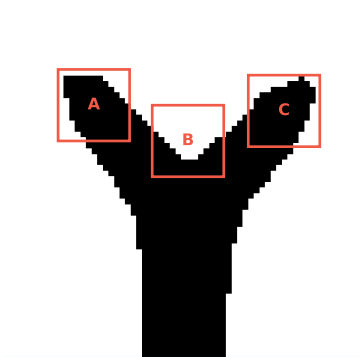
Gradient • Tmax=0.1533



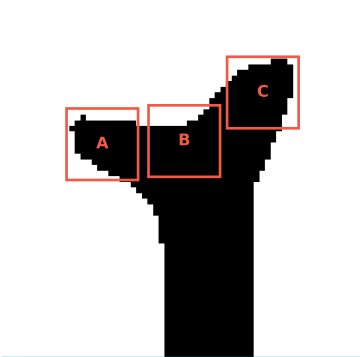
Gradient • Tmax=0.1588



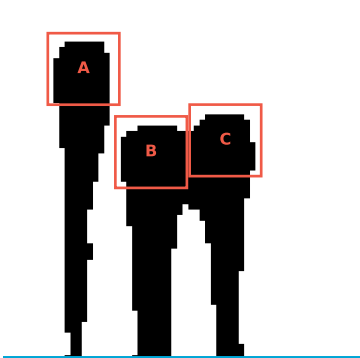
Gradient • Tmax=0.1683



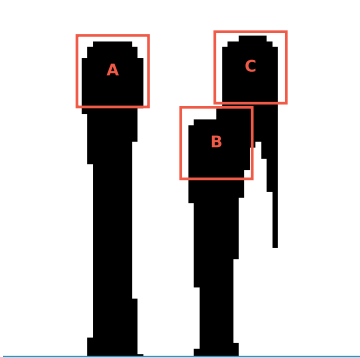
Gradient • Tmax=0.1504



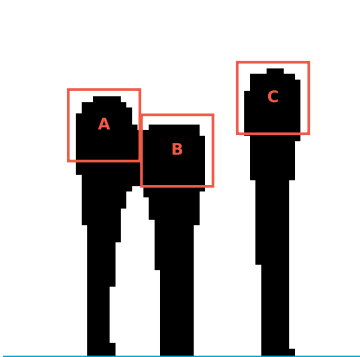
RAW • gap=47.2%



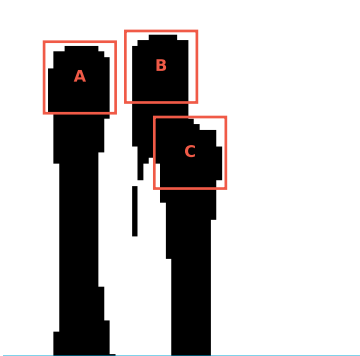
RAW • gap=14.3%



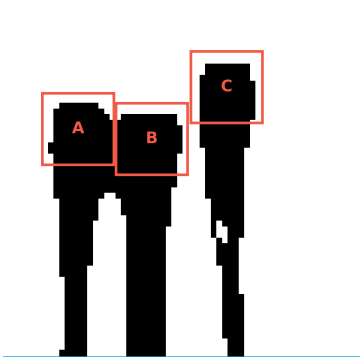
RAW • gap=39.5%



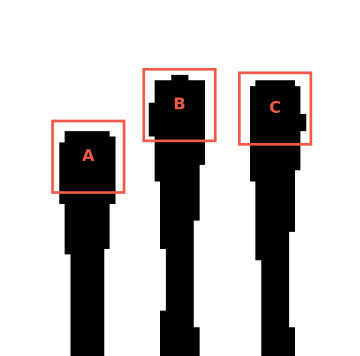
RAW • gap=18.7%



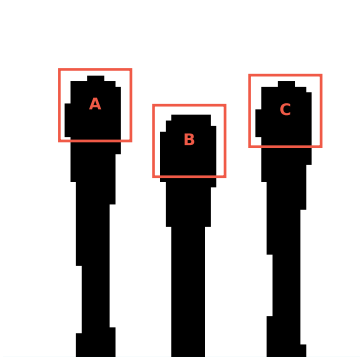
RAW • gap=40.7%



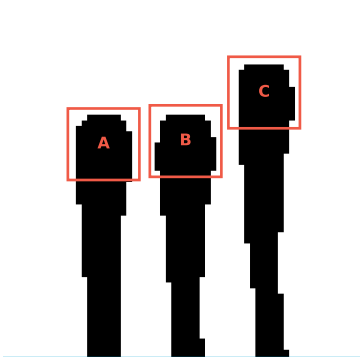
RAW • gap=27.2%



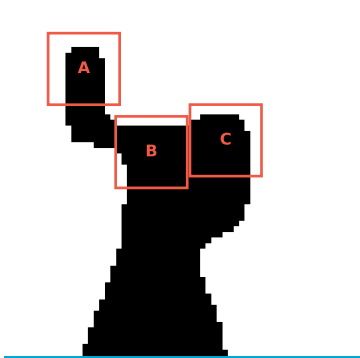
RAW • gap=24.3%



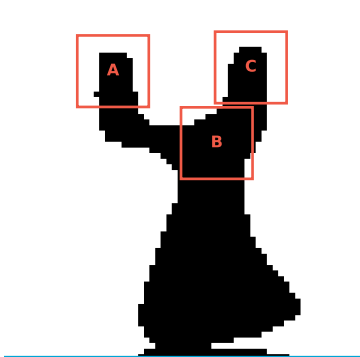
RAW • gap=36.9%



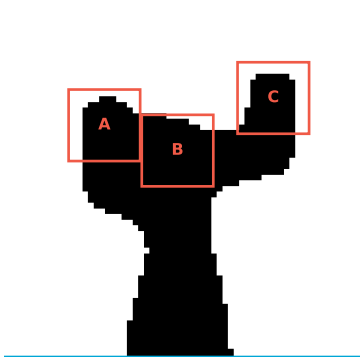
PHYSICS • gap=50.9%



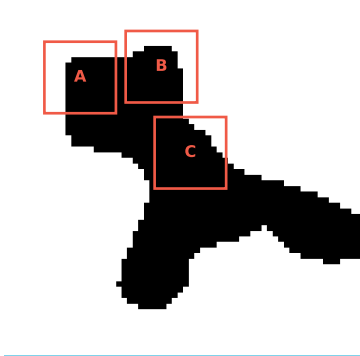
PHYSICS • gap=42.5%



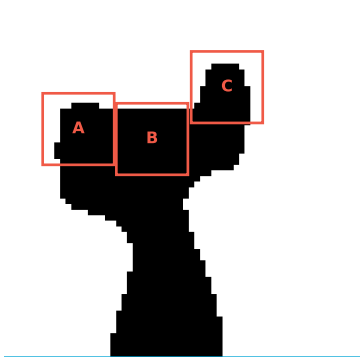
PHYSICS • gap=30.3%



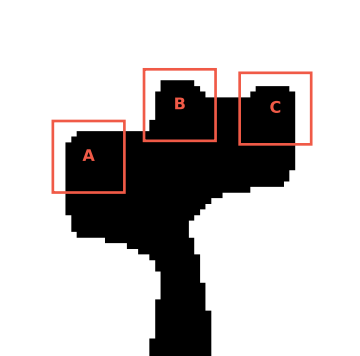
PHYSICS • gap=153.3%



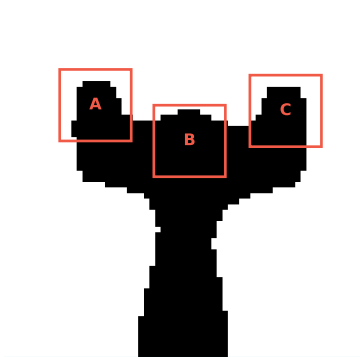
PHYSICS • gap=25.8%



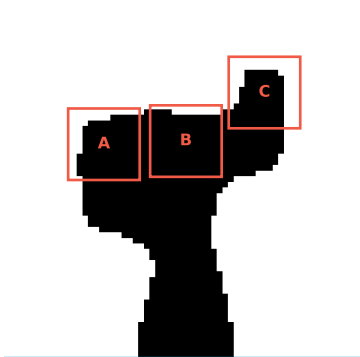
PHYSICS • gap=35.6%



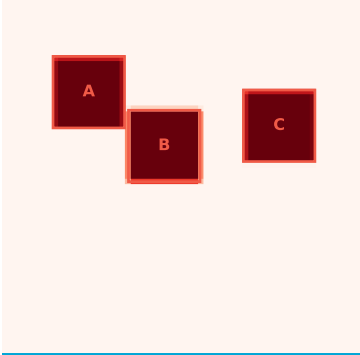
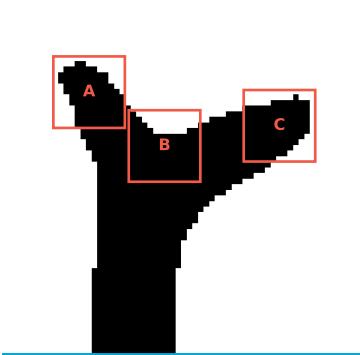
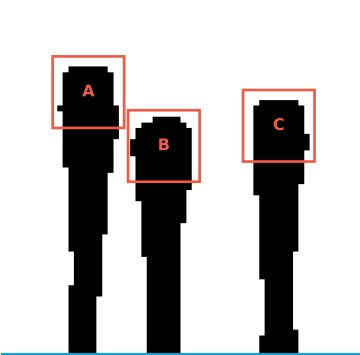
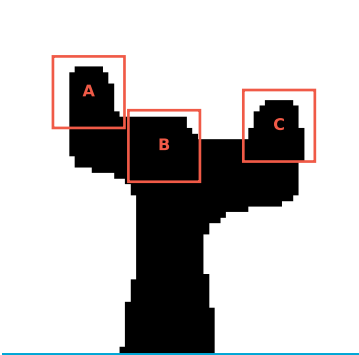
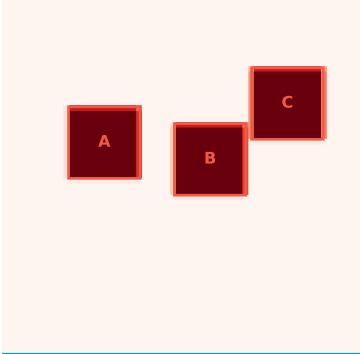
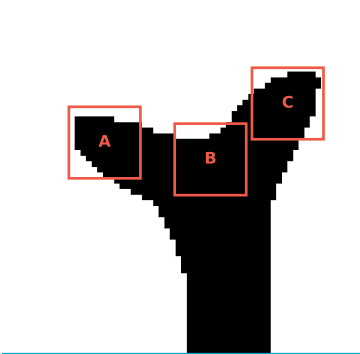
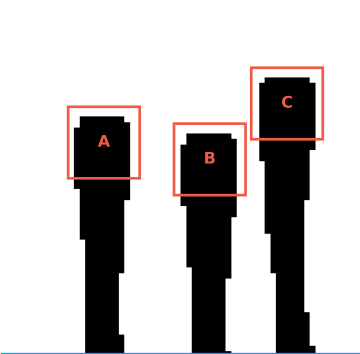
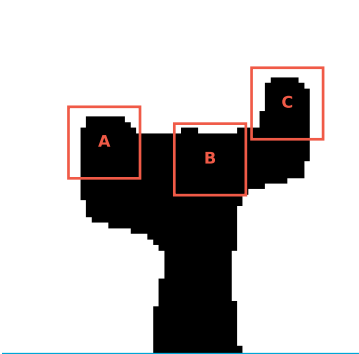
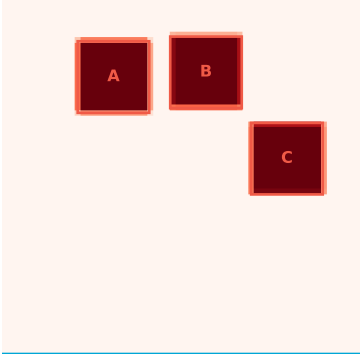
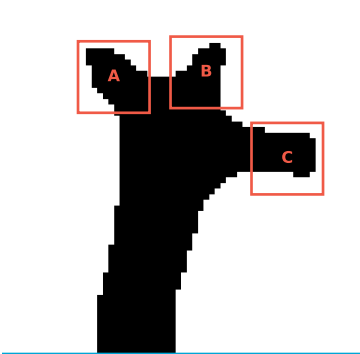
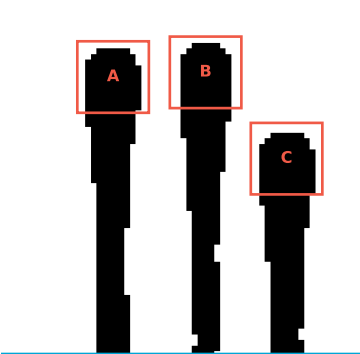
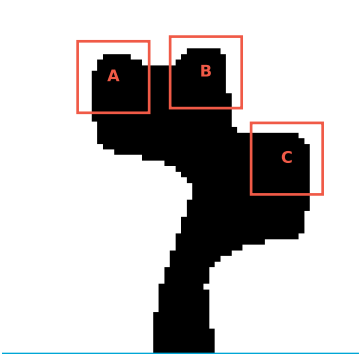
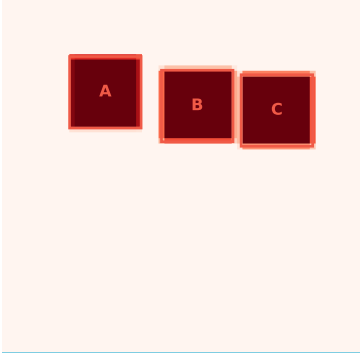
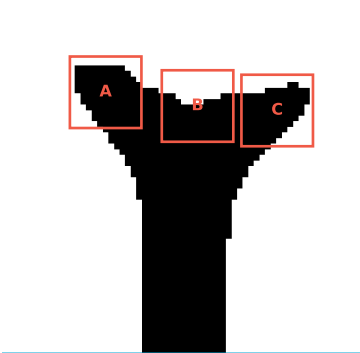
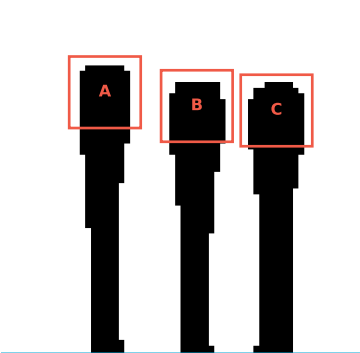
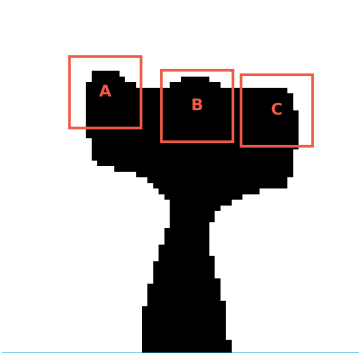
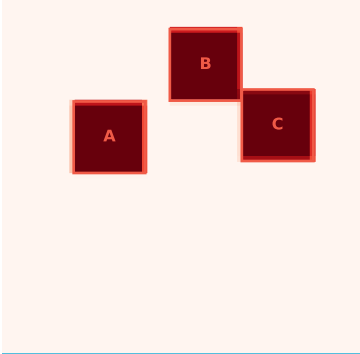
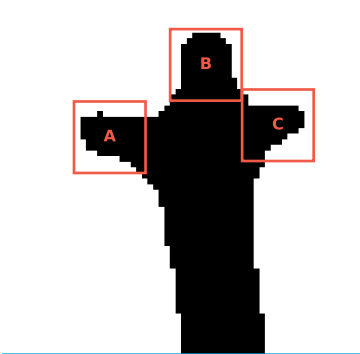
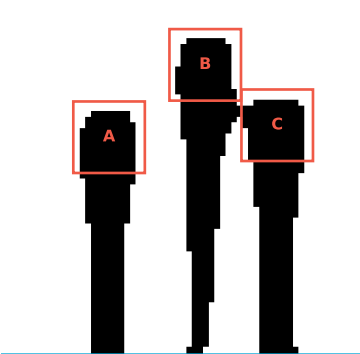
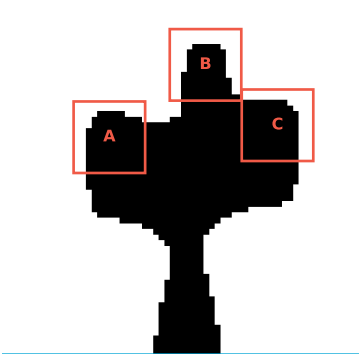
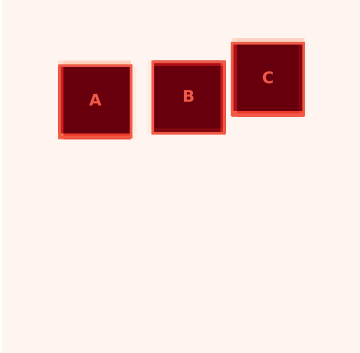
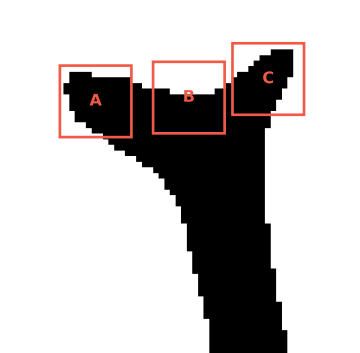
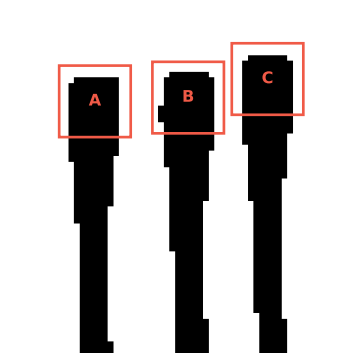
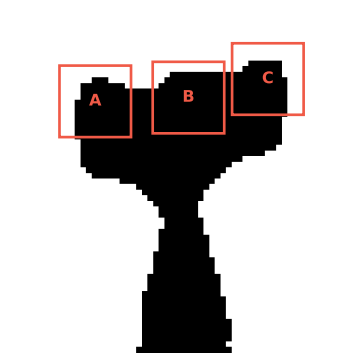
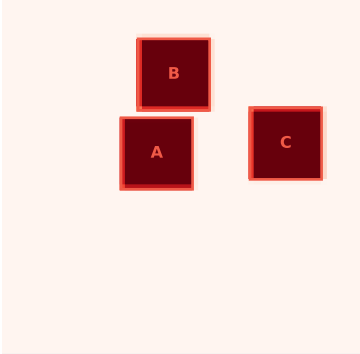
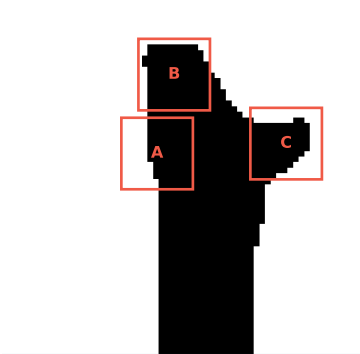
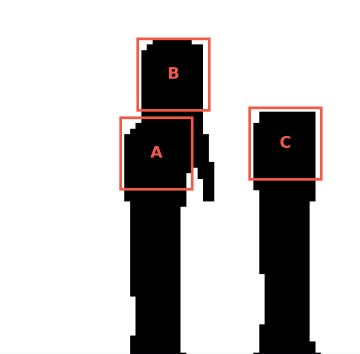
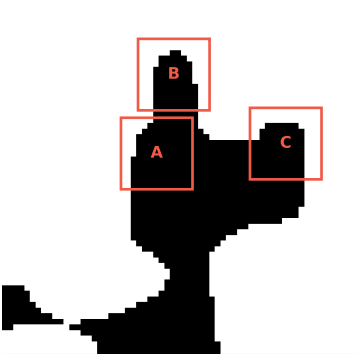
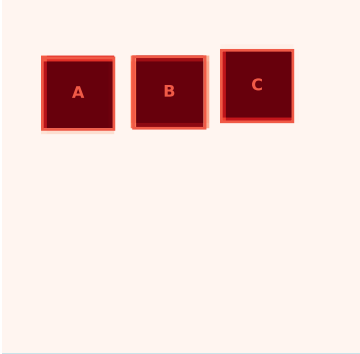
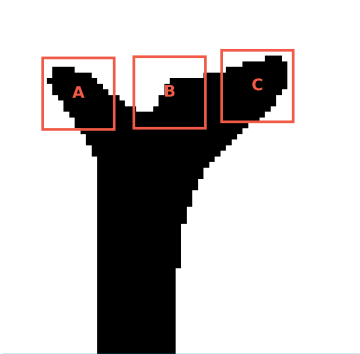
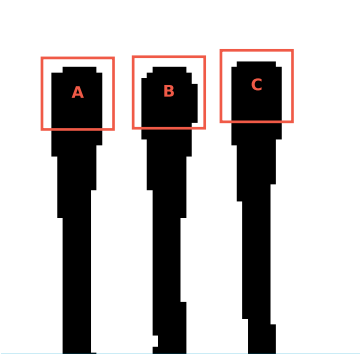
PHYSICS • gap=18.7%



PHYSICS • gap=33.0%



Complete validation topology atlas • layouts 24-31

Layout 24 • wide horizontal 	Gradient • Tmax=0.1708 	RAW • gap=18.1% 	PHYSICS • gap=16.8% 
Layout 25 • wide horizontal 	Gradient • Tmax=0.1514 	RAW • gap=27.7% 	PHYSICS • gap=28.5% 
Layout 26 • mixed 	Gradient • Tmax=0.1801 	RAW • gap=23.3% 	PHYSICS • gap=30.0% 
Layout 27 • wide horizontal 	Gradient • Tmax=0.1637 	RAW • gap=31.1% 	PHYSICS • gap=29.8% 
Layout 28 • wide horizontal 	Gradient • Tmax=0.1628 	RAW • gap=14.1% 	PHYSICS • gap=35.8% 
Layout 29 • wide horizontal 	Gradient • Tmax=0.1840 	RAW • gap=21.4% 	PHYSICS • gap=25.6% 
Layout 30 • vertically spread 	Gradient • Tmax=0.1427 	RAW • gap=20.4% 	PHYSICS • gap=44.9% 
Layout 31 • wide horizontal 	Gradient • Tmax=0.1824 	RAW • gap=22.5% 	PHYSICS • gap=28.0% 